

Predictive simulation of healthy and pathological gait with SCONE

BIOENG-404 Analysis and Modelling of Locomotion, EPFL. SCONE assignment (2021 handout)

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Setup

Model. `Human0914.osim` is a planar OpenSim 3 model with 9 degrees of freedom (pelvis tilt, horizontal and vertical pelvis translation, hip, knee and ankle on both legs) and 14 Hill-type muscles (hamstrings, gluteus maximus, iliopsoas, vasti, gastrocnemius, soleus and tibialis anterior on each side). Contact spheres at the heel and toes produce the ground reaction forces.

Controller. `ControllerComplexGH.scone` is the Geyer and Herr (2010) reflex controller with extra velocity (V_+) reflexes on soleus and gastrocnemius. A state machine detects five gait phases per leg (early stance, late stance, liftoff, swing, landing), and in each phase a set of reflexes of the form $u_m(t) = c_0 + k_l(l_m(t - \Delta t) - l_0) + k_v v_m(t - \Delta t) + k_f f_m(t - \Delta t)$ sets the muscle excitations. The 40 design parameters are reflex gains and offsets plus initial joint angle offsets.

Objective. `MeasureGait.scone` adds up five measures:

Measure	Weight	Penalizes
Gait	100	steps slower than <code>min_velocity</code> , and falling: the simulation stops when the centre of mass drops below 85 % of its initial height and the remaining time counts as standing still. Values below 0.05 count as 0
Effort	0.1	metabolic cost of transport (Wang et al. 2012), in J/(kg m)
DofLimits	0.1, 0.01	ankle outside [-60, 60] deg; knee limit torque above 5 Nm
HeadStabilityY	0.25	vertical head acceleration beyond 0.5 g
HeadStabilityX	0.25	horizontal head acceleration beyond 0.25 g

The gait term is roughly the fraction of the target speed that is missed, averaged over the steps, so a model that walks the full 10 s at or above the minimum speed scores 0 there. With the default weights, a good healthy gait is dominated by the effort term.

Optimization. SCONE runs a 10 s forward simulation per candidate and optimizes the parameters with CMA-ES (15 candidates per generation, 7 parents), starting from `InitParameters.par`.

How the simulations were run. The handout expects SCONE Studio on Windows. Here the official Linux build of SCONE 2.4.5 (OpenSim 3.3 backend) runs headless in Docker through its command line tool, so that every

run can be repeated from a terminal (scripts/scone.sh). The settings are identical to the GUI. Evaluating a solution prints the objective broken down into its measures, which is reported below for each best solution.

Gait analysis. SCONE Studio's Gait Analysis window was reimplemented in Python (src/scone_gait), following its source code: gait cycles start at foot contact (vertical ground reaction force above 0.001 body weight, stance of at least 0.1 s), the first two and last cycles are dropped, every cycle is resampled over 0 to 100 percent, and the mean curve is compared with the normative band of SCONE Studio. The fit of one plot is $100(1 - e_i/w_i)$, where e_i is how far the mean lies outside the band at the i -th normative sample and w_i the band width; the overall fit is the average over pelvis, hip, knee, ankle and ground reaction force. A foot contact index locates the centre of pressure at initial contact along the foot, from 0 at the heel to 1 at the metatarsal heads, to tell heel strikes from forefoot strikes. Signs follow the SCONE plots: hip and knee flexion positive, ankle dorsiflexion positive and plantarflexion negative.

Deliverable 1: healthy gait

Please comment on the results obtained from the values of the objective functions and gait analysis tool after evaluating your solution. What can be improved in terms of gait properties (please elaborate)?

Optimization. HealthyGait.scone was optimized for 50 generations. The model first completed 10 s of walking at generation 14 (objective 1.15), passed the 0.9 target at generation 20 (0.861), and reached its best value, **0.780**, at generation 35. The last 15 generations brought no further improvement, so the run had converged for this seed. The best solution is results/healthy/0035_1.021_0.780.par.

Measure	Weighted value	Raw value
Gait	0	step velocity 1.03 m/s over 17 steps (minimum 1.0)
Effort	0.722	cost of transport 7.22 J/(kg m)
DofLimits	0	knee limit torque 2.7 (left) and 2.8 Nm (right), below the 5 Nm threshold; ankle within range
HeadStabilityY	0.022	0.088
HeadStabilityX	0.036	0.146
Total	0.780	

Objective. The model walks the whole 10 s slightly above the required speed, so the gait term is zero, and 93 % of the objective is metabolic effort. The cost of transport of 7.2 J/(kg m) is about twice the gross metabolic cost measured in people walking at this speed (roughly 3 to 4 J/(kg m)), so the gait is stable but not efficient. The knee leans on its extension limit (2.7 to 2.8 Nm of limit torque on every stance), which is tolerated only because the joint limit measure ignores torques below 5 Nm. Head accelerations stay small.

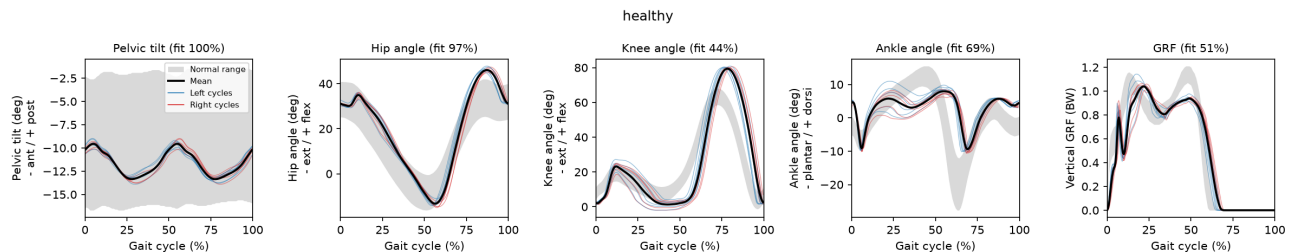


Figure 1: Gait analysis of the healthy solution. Grey: normal range from SCONE Studio. Thin lines: individual left (blue) and right (red) cycles. Black: mean of 11 cycles.

Gait analysis. Over 11 cycles the model walks at 1.01 m/s with a stride of 1.26 m, a cadence of 96 steps/min and a stance phase of 66 % of the cycle. The overall fit with the normal data is 72 %:

- **Pelvis and hip** are almost normal (fit 100 % and 97 %).
- **Knee** (fit 44 %): after a normal loading flexion of about 22 deg, the knee goes straight (0.7 deg at its least flexed) for the rest of stance, where people keep 5 to 10 deg of flexion. In swing it flexes to 80 deg instead of about 63 deg.
- **Ankle** (fit 69 %): the ankle plantarflexes quickly just after heel strike (foot slap, about -8 deg at 5 % of the cycle), dorsiflexes normally in midstance (peak 8 deg), but push-off is weak: the ankle only reaches -10 deg of plantarflexion, late (around 68 %), where people reach about -20 deg at 62 %.
- **Ground reaction force** (fit 51 %): a sharp impact spike at heel strike, then a dip to about 0.5 body weight, and a second peak of 0.95 instead of about 1.1 body weight.

What can be improved.

1. *Push-off.* The weak plantarflexion and low second force peak show that the plantarflexors contribute little to propulsion; the long stance and the large swing knee and hip flexion suggest that the model compensates by pulling the leg forward from the hip. A stronger push-off would shorten stance and probably lower the cost of transport.
2. *Loading response.* The impact spike, the foot slap and the dip in the force come from a heel strike that is not cushioned: in people the tibialis anterior lowers the foot eccentrically and the knee flexes under load. Penalizing high vertical forces (SCONE's own tutorials add a measure on ground reaction forces above 1.5 body weight) would push the optimizer towards a softer landing.
3. *Knee in stance.* The straight, limit-loaded knee in midstance is not physiological. The joint limit measure could penalize any limit torque (threshold 0 instead of 5 Nm), or a small knee flexion target could be added.
4. *Swing knee flexion* is about 17 deg too large, which costs energy and reflects the hip driven gait above.
5. *Optimization.* CMA-ES finds a local optimum that depends on the initial guess and the random seed. Longer runs, restarts from different seeds and more realistic objectives (for example a target speed of 1.2 to 1.3 m/s, closer to preferred walking speed) would give a better gait. The model itself is limited too: it is planar, its foot is a single rigid segment without a toe joint, and the reflex controller has no feedforward component.

Deliverable 2: heel walking from plantarflexor weakness

Please comment on the results obtained from the values of the objective functions and gait analysis tool after evaluating your solution. What are the main kinematic adaptations when the plantar flexors are weakened (please elaborate)?

Setup. Weakness.scone is HealthyGait.scone with the maximum isometric force of soleus and gastrocnemius scaled on both legs and with Measure05.scone (minimum speed 0.5 m/s). Three factors were tried with the handout method (start from InitParameters.par, at most 200 generations):

Factor	Outcome
0.25	never walked: best objective about 94 after 60 generations (the model falls in the first steps), stopped
0.5	walked from generation 176, best 1.196 at generation 198. Chosen value
0.7	walked from generation 18, best 0.965 at generation 99

As a check that the pattern does not depend on the optimizer's path, factor 0.5 was also optimized from the healthy solution (warm start, 112 generations, best 0.880). This is a deviation from the handout and is only used for comparison.

Measure	Healthy	x 0.7	x 0.5	x 0.5 warm
Gait	0	0	0	0
Effort (cost of transport, J/(kg m))	0.722 (7.22)	0.754 (7.54)	0.852 (8.52)	0.801 (8.01)
DofLimits (knee limit torque, Nm)	0 (2.7)	0.129 (6.7)	0.187 (9.4)	0 (2.9)
HeadStabilityY	0.022	0.031	0.093	0.032
HeadStabilityX	0.036	0.051	0.064	0.047
Total	0.780	0.965	1.196	0.880
Step velocity (m/s)	1.03	0.87	0.63	0.85

Objective. The weakened model meets the lower speed requirement (0.63 m/s against 0.5 m/s), so the gait term stays at zero, but every other term gets worse. Walking costs 18 % more energy per metre than in health even though it is slower, the knee now presses against its extension limit hard enough to be penalized (9.4 Nm, above the 5 Nm threshold), and vertical head accelerations are four times larger. The convergence is the most telling number: the weakened model needed 176 generations to find any gait that does not fall, against 14 in health, and a quarter of the normal strength was not enough at all. Plantarflexors are central to balance in this controller: the soleus force reflex is what stops the shank from rotating forward over the foot in stance.

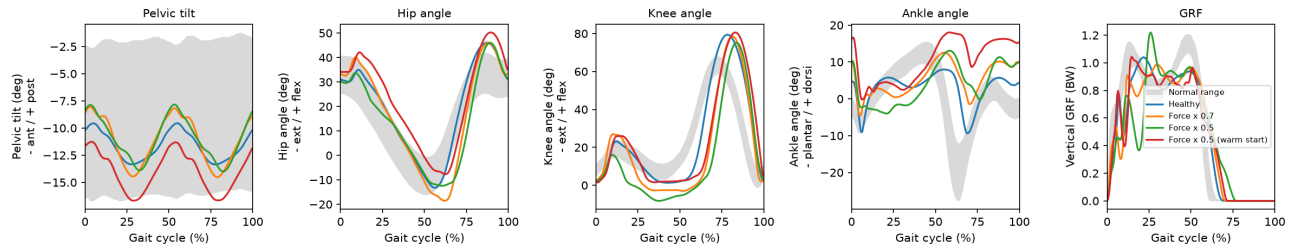


Figure 2: Mean gait cycles of the weakened models against the healthy solution. Grey: normal range.

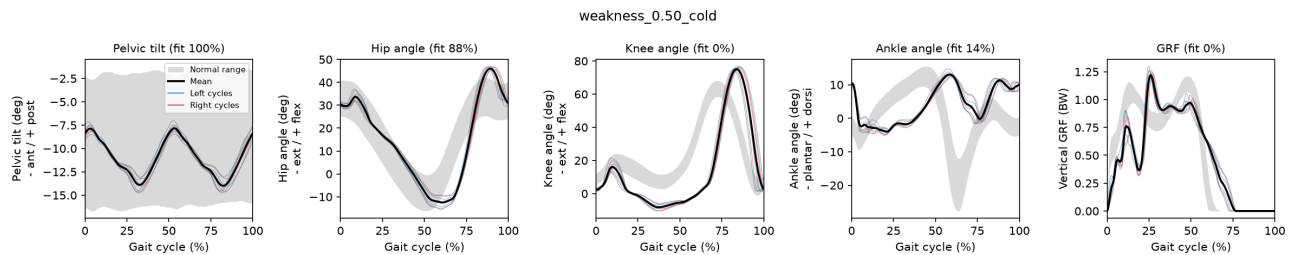


Figure 3: Gait analysis of the plantarflexor weakness x 0.5 solution (weakness.scone).

Metric	Healthy	x 0.7	x 0.5	x 0.5 warm
Speed (m/s)	1.01	0.86	0.65	0.84
Stride length (m)	1.26	1.28	1.08	1.14
Cadence (steps/min)	96	80	72	89
Stance (% of cycle)	66	69	75	71
Ankle at contact (deg)	4.6	9.8	10.1	16.1
Peak dorsiflexion in stance (deg)	8.4	12.6	13.2	18.0
Peak plantarflexion (deg)	-10.2	-4.6	-4.5	-0.2
Knee, least flexed in stance (deg)	0.7	-3.2	-8.3	1.4
Foot contact index	0.00	-0.02	-0.01	-0.05
Overall fit (%)	72	45	40	37

Kinematic adaptations. The weakened model walks on its heels:

1. *Excessive dorsiflexion.* The foot lands on the heel (contact index near zero) with the ankle dorsiflexed by 10 deg instead of 5 deg, and the shank keeps rotating forward over the foot through stance, up to 13 deg of dorsiflexion (18 deg with the warm start) where the healthy model stops at 8 deg. Weak plantarflexors cannot brake the forward rotation of the tibia in midstance and late stance.
2. *No push-off.* Healthy late stance ends with a quick plantarflexion to -10 deg; with half the strength the ankle barely passes neutral (-4.5 deg, -0.2 deg with the warm start), so the “ankle rocker” that propels the body is lost. The second ground reaction force peak is replaced by a long, flat load, and stance stretches to 75 % of the cycle.
3. *Slow, short, careful steps.* Speed falls from 1.01 to 0.65 m/s, mostly through cadence (96 to 72 steps/min) and stride length (1.26 to 1.08 m).
4. *A knee strategy.* Without the soleus to hold the shank, the cold solution locks the knee in hyperextension through stance (-8 deg), letting the passive knee limit carry the load. The warm-started solution keeps the knee near straight but dorsiflexes even more and extends the hip less. Both are compensations for the missing ankle moment, and they show that the optimizer can land in different local optima for the same impairment.

The effects grow with the weakness: x 0.7 already shows every adaptation in a milder form. These patterns match the calcaneal gait seen after plantarflexor weakness, for example after over-lengthening of the Achilles tendon in cerebral palsy, where excessive stance dorsiflexion and lost push-off are the hallmark. Clinically the knee usually ends up flexed (crouch) rather than hyperextended; the model's knee hyperextension is allowed by its soft knee limit and by an objective that tolerates limit torque.

Deliverable 3: toe walking from hyperreflexia

Please comment on the results obtained from the values of the objective functions and gait analysis tool after evaluating your solution. What are the main kinematic adaptations when hyperreflexia is introduced to the plantar flexors (please elaborate)?

Setup and one deviation. Hyperreflexia.scone uses ControllerHyperreflexia.scone, a copy of the course controller in which the stance velocity (V+) reflex gain of soleus and gastrocnemius is raised from its default of 0.1, and Measure05.scone. The handout writes the new gain as $KV = \sim 1.0 < 0, 10 >$. The tilde keeps KV a design parameter, so CMA-ES starts at 1.0 but may move it anywhere in [0, 10]. In the healthy optimization these two gains fell from 0.1 to below 0.04, so the optimizer clearly prefers to switch this reflex off. A spastic patient cannot do that. The chosen solution therefore uses a **fixed gain**, $KV = 1.0$ without the tilde, the same choice the SCONE hyper-reflexia tutorial makes. The handout version was run too, for comparison.

Variant	First full walk	Best objective (generation)
KV = 0.3, fixed	generation 8	1.330 (99)
KV = 1.0, fixed (chosen)	generation 11	1.227 (101)
KV = $\sim 1.0 < 0, 10 >$ (handout, optimized)	generation 14	1.295 (107)

The runs were stopped after about 100 generations, when the objective was improving by less than 0.2 % per generation.

Measure	Healthy	KV 0.3	KV 1.0	Handout
Gait	0	0	0	0
Effort (cost of transport, J/(kg m))	0.722 (7.22)	1.118 (11.18)	1.029 (10.29)	1.046 (10.46)
DofLimits	0	0	0	0
HeadStabilityY	0.022	0.031	0.014	0.009
HeadStabilityX	0.036	0.181	0.184	0.241
Total	0.780	1.330	1.227	1.295

Measure	Healthy	KV 0.3	KV 1.0	Handout
Step velocity (m/s)	1.03	0.70	0.71	0.74

Objective. The toe walker meets the 0.5 m/s requirement and respects the joint limits, but walking costs 42 % more energy per metre than in health (10.3 against 7.2 J/(kg m)) at a lower speed: without a heel rocker the plantarflexors have to hold the body up through the whole stance. Fore-aft head accelerations are five times larger (0.184 against 0.036): every step lands on a stiff, plantarflexed foot and brakes the body abruptly. The vertical head term is smaller than in health because the heel impact spike is gone.

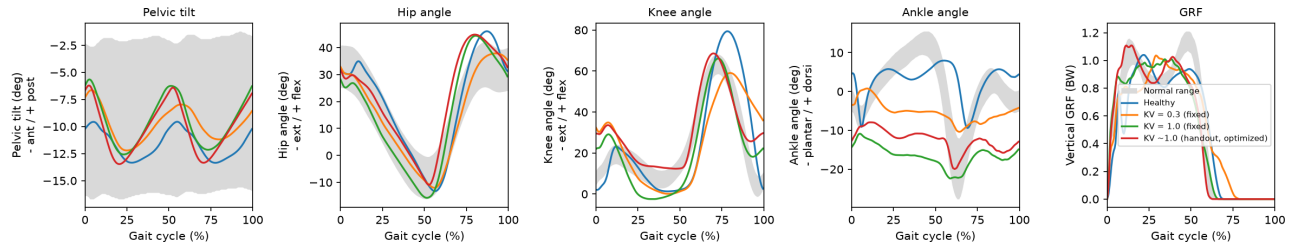


Figure 4: Mean gait cycles with plantarflexor hyperreflexia against the healthy solution. Grey: normal range.

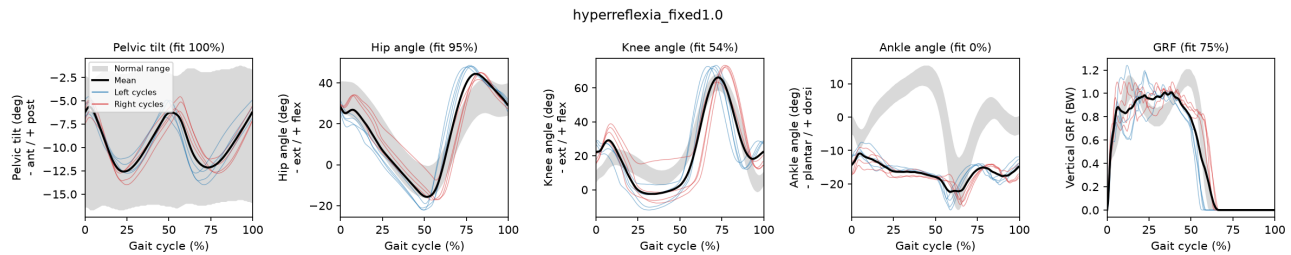


Figure 5: Gait analysis of the hyperreflexia solution with KV = 1.0 (Hyperreflexia.scone).

Metric	Healthy	KV 0.3	KV 1.0	Handout
Speed (m/s)	1.01	0.67	0.73	0.71
Stride length (m)	1.26	0.88	1.06	1.00
Cadence (steps/min)	96	92	82	86
Stance (% of cycle)	66	68	61	60
Ankle at contact (deg)	4.6	-3.6	-14.4	-12.7
Peak dorsiflexion in stance (deg)	8.4	1.7	-10.8	-7.8
Peak plantarflexion (deg)	-10.2	-14.6	-25.9	-20.7
Knee at contact (deg)	2.1	32.4	22.0	29.5
Knee, least flexed in stance (deg)	0.7	-1.3	-2.9	12.1
Peak knee flexion in swing (deg)	80.1	77.0	71.5	69.2
Foot contact index	0.00	1.23	1.26	1.27
Overall fit (%)	72	45	65	67

Kinematic adaptations. With KV = 1.0 the model walks on its toes:

1. *Forefoot contact and no heel contact.* The first contact is on the toes (contact index 1.26, beyond the metatarsal heads) with the ankle plantarflexed by 14 deg, and the heel never reaches the ground: even the most dorsiflexed point of stance is still 11 deg of plantarflexion, where the healthy model reaches 8 deg of dorsiflexion. The whole ankle curve is shifted by 15 to 20 deg towards plantarflexion (ankle fit 0 %), with a peak of -26 deg at push-off.

2. *Why.* A velocity reflex fires whenever soleus and gastrocnemius are being stretched, which in a normal stance happens all the time, as the shank rotates forward over the planted foot. With a high gain, every attempt to dorsiflex is met by a strong plantarflexor contraction, so the optimizer settles on a gait that never stretches these muscles quickly: land already plantarflexed and stay on the forefoot.
3. *Knee.* The knee lands flexed (22 deg instead of 2 deg) to absorb the forefoot landing, then is pushed into slight hyperextension in midstance (-3 deg): with the foot fixed in plantarflexion, the ground reaction force passes in front of the knee (the plantarflexion and knee extension couple). Swing knee flexion is smaller (72 deg instead of 80 deg).
4. *Gait pattern.* Slower walking (0.73 m/s) with shorter strides, a lower cadence, and a single rounded force peak without the heel strike transient (GRF fit 75 %, higher than the healthy model's because the healthy impact spike is gone).

Effect of the gain. At $KV = 0.3$ the model already lands on its forefoot (contact index 1.23) but with the ankle near neutral, the heel close to the ground in midstance, and irregular cycles; at $KV = 1.0$ the toe walking is complete and regular. Toe walking therefore starts at or below three times the default gain.

The handout version also produces toe walking, but the optimizer reshaped the impairment: by generation 107 it had lowered the soleus gain to 0.44 and raised the gastrocnemius gain to 1.56. Gastrocnemius also flexes the knee, and that solution keeps the knee flexed through stance (at least 12 deg), similar to the “jump gait” of children with cerebral palsy, who combine equinus with knee flexion. It is an interesting gait, but the impairment level is chosen by the optimizer rather than set by the modeller, which is why the fixed gain is used as the answer.

Deliverable 4: own model, toe walking from plantarflexor contracture

Propose a biomechanical or neural model to reproduce heel or toe walking. You can modify another biomechanical or a neural parameter similarly to previous questions. Explain why you expect your model to result in a pathological gait. [...] Please comment on the results obtained from the values of the objective functions and gait analysis tool after evaluating your solution. If your solution is not satisfying, make a hypothesis regarding eventual biomechanics or neural compensations.

Model. Deliverable 3 produced toe walking with a *neural* impairment. The proposed model produces it with a *biomechanical* one: a contracture of the plantarflexors, as in the equinus of children with cerebral palsy, where the triceps surae and Achilles tendon are too short. In `Model.scone` the tendon slack length of soleus and gastrocnemius is shortened on both legs (`tendon_slack_length.factor`); the controller is unchanged, so no `ControllerModel.scone` is needed.

Why it should cause toe walking. The length of a muscle-tendon unit is set by the joint angles. If the tendon is shorter, the muscle fibers must be longer at every ankle angle, so the passive elastic element of the muscle starts pulling at a more plantarflexed angle and pulls harder at any given dorsiflexion. This creates a passive plantarflexion moment that grows as the ankle dorsiflexes and does not depend on neural control. In this model, shortening the tendon by 5 % stretches the soleus fibers by 25 % of their optimal length (gastrocnemius 20 %), and by 10 % stretches them by 50 % (gastrocnemius 40 %). The foot should therefore not reach a plantigrade position at contact, and the heel should stay off the ground once the passive moment exceeds what body weight can overcome.

Tendon slack factor	First full walk	Best objective (generation)
0.95	generation 3	1.146 (100)
0.90 (chosen)	generation 0	1.014 (150)

Measure	Healthy	x 0.95	x 0.90	Hyperreflexia KV 1.0
Gait	0	0	0	0

Measure	Healthy	x 0.95	x 0.90	Hyperreflexia KV 1.0
Effort (cost of transport, J/(kg m))	0.722 (7.22)	0.955 (9.55)	0.904 (9.04)	1.029 (10.29)
DofLimits	0	0	0	0
HeadStabilityY	0.022	0.013	0.002	0.014
HeadStabilityX	0.036	0.178	0.108	0.184
Total	0.780	1.146	1.014	1.227

Objective. Both contractures walk at 0.8 m/s without breaking any joint limit. The cost of transport rises by 25 to 32 % compared with health, less than with hyperreflexia (42 %), and fore-aft head accelerations are three to five times larger, again from landing on a plantarflexed foot. Walking was found almost at once (generation 0 to 3): a passive constraint is easier for the optimizer to work with than an active reflex that fights every stretch.

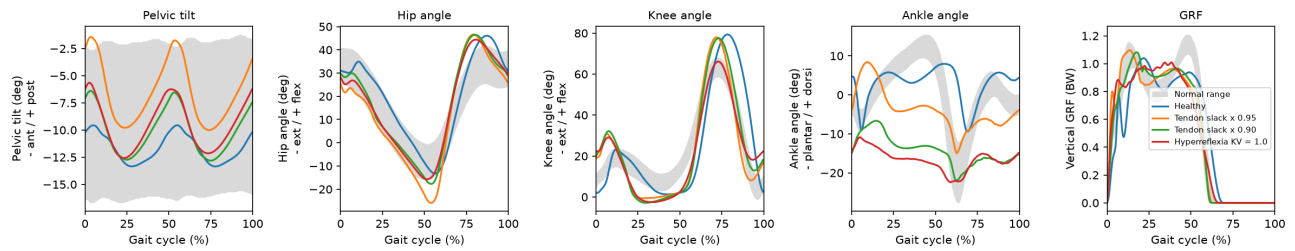


Figure 6: Mean gait cycles with plantarflexor contracture, against the healthy solution and the hyperreflexia solution of Deliverable 3. Grey: normal range.

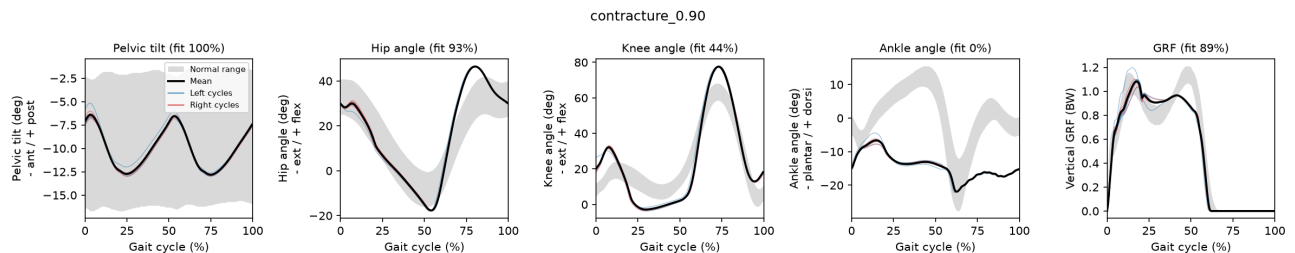


Figure 7: Gait analysis of the plantarflexor contracture solution with tendon slack x 0.90 (Model.scone).

Metric	Healthy	x 0.95	x 0.90	Hyperreflexia KV 1.0
Speed (m/s)	1.01	0.78	0.82	0.73
Cadence (steps/min)	96	81	87	82
Ankle at contact (deg)	4.6	-3.8	-15.0	-14.4
Peak dorsiflexion in stance (deg)	8.4	8.5	-6.6	-10.8
Peak plantarflexion (deg)	-10.2	-15.6	-22.0	-25.9
Knee at contact (deg)	2.1	18.6	19.9	22.0
Knee, least flexed in stance (deg)	0.7	-0.6	-2.9	-2.9
Peak hip extension (deg)	-13.7	-26.1	-17.8	-18.0
Foot contact index	0.00	1.04	1.26	1.26
Overall fit (%)	72	68	65	65

Results. The model reproduces toe walking, and its severity follows the size of the contracture:

- With **x 0.90** the gait is full toe walking: contact on the toes (index 1.26) with the ankle plantarflexed by 15 deg, the heel never reaches the ground (stance never gets past 7 deg of plantarflexion), and push-off

reaches -22 deg. The knee lands flexed (20 deg) and is pushed into slight hyperextension in midstance, the same plantarflexion and knee extension couple as with hyperreflexia.

- With x 0.95 the model lands on the ball of the foot (index 1.04, ankle -4 deg) and then lowers its heel: the ankle dorsiflexes to 8.5 deg in midstance before an early, strong push-off. This “toe-heel” pattern is typical of a mild equinus.

Contracture or spasticity? The kinematics of the two causes are close, but not identical. With the contracture the ankle still dorsiflexes slowly in early stance (from -15 deg at contact to about -7 deg) because passive tension depends on length, not on speed; with hyperreflexia the ankle stays almost flat, because any fast stretch is answered by a contraction. This is the difference clinicians probe when they move the ankle slowly and then quickly during an examination (the Tardieu scale), and it shows why the cause of a toe walking pattern cannot easily be read from gait kinematics alone.

Compensations. The x 0.90 solution is satisfying. For the milder x 0.95 contracture, the heel does come down, and the solution suggests how: the reflex gains of the plantarflexors are almost unchanged from the healthy solution (soleus force gain 0.27 to 0.32), so the heel is not lowered by switching off the plantarflexors. Instead body weight stretches the contracted muscles, helped by a knee that locks straight in midstance and by a much larger hip extension at the end of stance (-26 deg against -14 deg), which carries the body over the stiff ankle. In a patient, further compensations would be possible that this model cannot produce, for example stronger tibialis anterior activity in swing to lift the forefoot, or reduced spinal excitability; a feedforward component in the controller would be needed to test them.

Extension: can crouch gait emerge?

This section is not part of the 2021 handout. Crouch gait, walking with excessive knee and hip flexion throughout stance, is one of the most common gait patterns in ambulatory children with cerebral palsy. Short hamstrings and hip flexors are often blamed for it, so knee and hip flexor contractures were modelled the same way as in Deliverable 4.

Model	Start	Outcome
Hamstrings, tendon slack x 0.90	handout initial guess	never walked (best 89.8 after 145 generations), stopped
Hamstrings, tendon slack x 0.90	healthy solution	never completed 10 s in 200 generations (best 50.2)
Hamstrings, tendon slack x 0.95	healthy solution	walked, best 0.958
Hamstrings and iliopsoas, tendon slack x 0.95	healthy solution	walked, best 0.859

The contracture runs were warm started from the healthy solution, since the cold start did not find a gait at all.

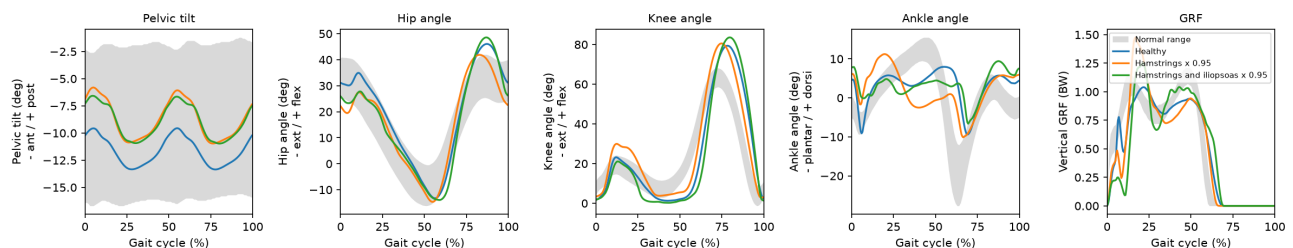


Figure 8: Mean gait cycles with knee and hip flexor contractures against the healthy solution. Grey: normal range.

No crouch emerged. The mild contractures change the gait only a little: the knee flexes more during loading with short hamstrings (30 deg against 23 deg), but in midstance it is as straight as in health (least flexed 3.1 deg

and 0.2 deg against 0.7 deg). The clearest change is at the pelvis: both models tilt it about 3 deg posteriorly on average (-8.7 and -8.9 deg against -11.7 deg). A posterior pelvic tilt slackens the hamstrings at the hip, so the knee does not have to stay flexed. This is the compensation seen in people with tight hamstrings. A stronger contracture made walking impossible instead of producing a crouch.

Why the model avoids crouch. Three reasons are likely:

1. *The objective.* Crouch needs large, sustained quadriceps forces to hold the flexed knee, so it costs much more energy than upright walking. An optimizer that minimizes metabolic cost will use any other compensation first (here the pelvis), and only accept a crouch if upright walking is impossible.
2. *What is missing from the model.* In children with cerebral palsy crouch usually comes with bone deformities (femoral anteversion, tibial torsion) that reduce the moment arms of the antigravity muscles, and with weakness and impaired selective control. The planar model has none of these. A predictive study of one child with crouch gait (Falisse et al. 2020) found that personalized muscle-tendon properties, on a model that included the child's bone deformities, were what reproduced the crouch, not reduced control complexity or spasticity.
3. *Hamstring length.* Shortening the hamstrings is not the whole story clinically either: in many children who walk in crouch, the hamstrings operate at normal or long lengths during gait (Arnold et al. 2006).

Testing crouch properly would need impairments that remove the upright option: combined plantarflexor weakness (in Deliverable 2 the model kept its knee straight only by hyperextending it against the joint limit), a stiffer knee extension limit, reduced quadriceps strength, or an objective that weighs effort less.

Summary

Deliverable	Impairment	Gait produced	Key signs
1	none	near-normal walking at 1.0 m/s, objective 0.780	stiff knee in midstance, weak push-off, heel impact spike
2	plantarflexor force x 0.5	heel walking at 0.65 m/s	dorsiflexed contact, excessive stance dorsiflexion, no push-off, knee hyperextension
3	stance V+ reflex gain of soleus and gastrocnemius fixed at 1.0	toe walking at 0.73 m/s	forefoot contact, heel never down, ankle 15 to 20 deg more plantarflexed
4	plantarflexor tendon slack length x 0.90	toe walking at 0.82 m/s	as Deliverable 3, but the ankle still dorsiflexes slowly in early stance
Extension	hamstring and iliopsoas contracture	no crouch; posterior pelvic tilt instead	

These results agree with a study by the group that wrote the assignment (Bruehl et al. 2022), which used an extended version of the same controller: there, too, plantarflexor hyperreflexia produced toe walking while muscle or neural weakness only partly produced heel walking. Here the weakness did produce a clear heel gait, but it was by far the hardest condition for the optimizer.

Limitations. Each condition was optimized once, from one initial guess and one random seed, and CMA-ES finds local optima: the two different knee strategies found for the same weakness in Deliverable 2 show how much this matters. Most pathological runs were stopped at about 100 generations, before the 200 allowed, when they had plateaued. The model is planar, its feet have no toe joint, the knee extension limit is soft, and the objective does not penalize knee hyperextension below 5 Nm of limit torque, which the optimizer used in several solutions. The gait analysis follows SCONE Studio's code but runs outside the GUI; its speed agrees with SCONE's own gait measure within 3 %.

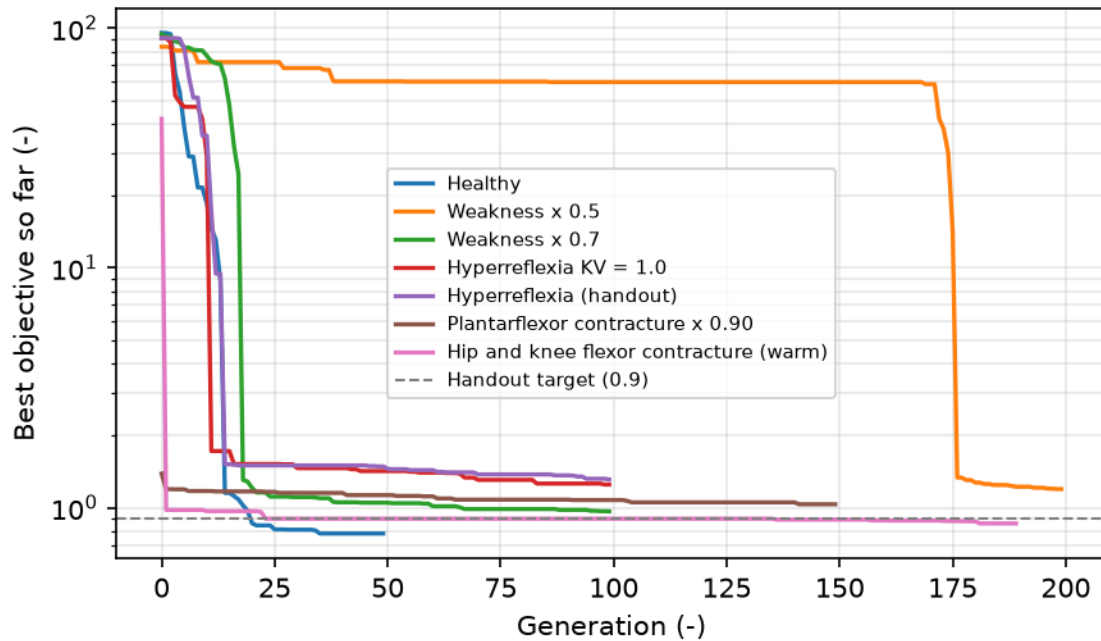


Figure 9: Best objective against generation for the chosen solutions. The weakened model needed 176 generations to find a gait that does not fall.

Reproducibility

All runs used SCONE 2.4.5-RC-2 (OpenSim 3.3 backend) from the official scone-studio Linux build, in Docker, on an Apple M5 laptop (10 cores, 16 GB RAM) under amd64 emulation. On the idle machine a 10 s simulation took 5 to 6 s and a generation of 15 walking candidates about 20 s; the 15 optimizations of this report (about 1,900 generations, 28,000 simulations) took about 12.5 hours of wall time with several runs in parallel (docs/compute.md has the full log). Thread count did not change the results: 3 and 10 threads gave identical optimization histories. Every optimized scenario is in `scone/sweeps/`, the handout-named scenarios are in `scone/`, and each curated result folder in `results/` contains the setup files SCONE copied, the best `.par` file, its evaluated `.par.sto` and the objective breakdown `.par.txt`. A solution can be replayed with `scripts/scone.sh evaluate results/<name>/<best>.par`. The notebook docs/notebook.md lists every run, including the failed ones, in order.

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